

Missing-Data Imputation With Position-Encoding Denoising Autoencoders for Industrial Processes

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Abstract—Missing values are a common occurrence in industrial datasets, resulting from the multiple sampling rates, sensor malfunctions, and transmission errors, whose presence can significantly affect the accuracy of data-driven models. An effective method to solve this problem is to impute the missing data in advance. This article proposes a new position-encoding denoising autoencoder (PE-DAE), which is motivated by the advantages of DAE in data reconstruction. To make use of the known information, the autocorrelation of the time-series data and the information on the missing positions are considered. Moreover, a self-paced learning (SPL) training strategy is proposed to improve the imputation performance under different levels of the missing data. The SPL training framework can first learn the knowledge structure of data with low missing rates and then gradually increase the difficulty, transitioning to learning more complex knowledge from data with higher missing rates. Finally, the proposed method is used for missing-data imputation in two real industrial processes. Comparative experiments show that the PE-DAE+SPL achieves the smallest error at all missing rates.

Index Terms—Deep learning (DL), denoising autoencoder (DAE), industrial processes, missing-data imputation.

I. INTRODUCTION

IN RECENT years, with the advent of the big-data era, data-driven process monitoring, fault diagnosis, fault classification, and soft sensors have become areas of strong research interest [1], [2], [3], [4]. However, the success of the data-driven methods depends heavily on the quality of the process data. Due to problems in data collection, data transmission, and data management, industrial data obtained usually have different degrees of missing values [5]. These

missing data not only increase the difficulty of data analysis but also cause errors in the data-driven models, which fail to provide proper guidance for industrial production [6], [7], [8].

There are two common strategies for handling samples with missing data: listwise deletion and data imputation [9]. The most straightforward method is to delete samples with missing data, leaving only complete samples, but this results in fewer available samples and loss of information [10]. Data imputation replaces the missing values with an estimate provided by the variable itself or other related variables [11]. Since data imputation does not reduce the available data, it has become the most popular method for processing missing data.

Many data imputation methods have been developed, including statistical methods such as simple imputation (mean, median, or mode values), low-rank matrix completion, and multiple imputations [12]. These methods provide preliminary estimates for individual variables without considering spatiotemporal correlations that may exist among them. Consequently, they are commonly used as initial values for missing data in machine-learning- and deep-learning-based imputation techniques. Notable examples of machine-learning-based imputation approaches include principal component analysis imputation [13], k -nearest neighbors imputation [14], soft impute [15], and support vector machines imputation [16]. These methods train the model with complete data and evaluate it on incomplete data. However, during the training process, nonmissing values in the incomplete samples are excluded. The omission of nonmissing values results in the loss of potentially valuable information, leading to reduced imputation performance [17]. Furthermore, the effectiveness of imputation is heavily influenced by the model's generalizability and its capability to extract complex features. Machine-learning-based methods, particularly when handling multivariate time-series data from complex systems, may face challenges in effectively extracting the valuable features [18], [19], [20], [21], [22].

Recently, deep learning (DL) has been widely used in data imputation due to its ability to handle strong nonlinearities and good data representation [23]. Typically, DL-based data imputation methods consist of two stages: model training and imputation. In the model-training stage, most methods use complete datasets to train the deep-learning models, but the networks trained in this way are usually difficult to generalize the situations with missing data. In the data-imputation phase, since deep networks cannot directly handle samples with missing values, constant filling is often used to preprocess incomplete samples. After obtaining the pre-

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imputation data, it is used as the network input for DL to calculate the corresponding network output as the final missing-data estimate [24], [25], [26]. At present, the mainstream DL-based data-imputation methods include denoising autoencoders (DAEs), generative adversarial networks [27], and convolutional neural networks [28]. Since the learning goal of the DAE is to reconstruct the repaired input, it is very suitable for recovering incomplete samples. In addition, the DAE can also form a deep network by stacking layer by layer, which improves its ability for complex data imputation. Therefore, the DAE-based methods have achieved great success in the field of missing-data imputation. Abiri et al. [29] developed a consistent framework for the missing data with multitype variables, which is based on a DAE. Lai et al. [30] proposed a tracking-removed autoencoder for the missing value imputation, which helps the neural network learn the cross correlation between variables. Wang et al. [31] tackled the issue of missing data in manufacturing processes by introducing an innovative autoencoder based on the orthogonal least squares. This autoencoder system individually selects the hidden neurons using orthogonal least-squares estimation until a sufficient network is established. Recent advancements in DL have been pivotal in tackling real-world industrial challenges characterized by imperfect data. Shen et al. [32] proposed a multiresolution pyramid variational autoencoder (MR-PVAE) model to address the issue of missing data caused by multirate in the data-driven soft sensor development for industrial processes. This model utilized deep feature extraction and feature pyramid augmentation, enabling the effective utilization of the process data with discrepant sampling rates. Liu et al. [33] proposed a novel prediction modeling method based on the transformer model, named Multirate-Former, to address the incomplete data caused by the multirate sampling in industrial processes. Cao et al. [34] proposed BRITS, which utilizes bidirectional recurrent neural networks to handle the missing values in time-series data and integrates a temporal decay mechanism to assess the significance of observed data points relative to the missing values. However, most of these methods initialize the missing values by replacing values at missing positions with the mean of nonmissing values for each variable. Subsequently, these samples are fed into a deep-learning model, making it challenging for the model to know which positions are missing. The model learns information from all the positions based on the mean values, which weakens the imputation effectiveness for the truly missing positions. Furthermore, most of these methods overlook the ramifications arising from disparities in the missing rates between real-world data and training data. The majority of these models are trained using training datasets with a fixed missing rate. However, the actual missing rate in real-world data is variable.

To estimate missing values more accurately, this article proposes an algorithm called position-encoding DAE (PE-DAE). The original DAE learns a function to recover the clean data from corrupted data. However, it does not consider variable autocorrelation and the information regarding the missing locations. Therefore, PE-DAE redesigns the network structure and training strategy of the neural network. First, the information regarding the missing position of the input data is coded to form a new positional encoding, which is

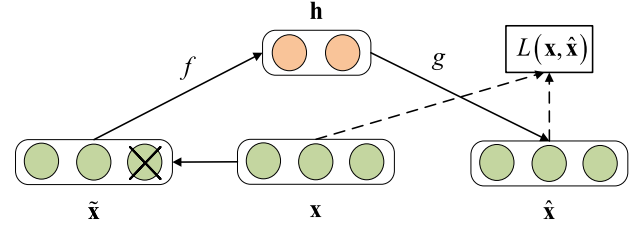


Fig. 1. Structure of a DAE.

used as input to the PE-DAE together with the original input variables. Thus, this approach allows the neural network to focus on the learning data completion ability for missing positions. Then, the input variables are dynamically expanded to consider the autocorrelation of the missing variables to help data imputation. Finally, a self-paced learning (SPL) training framework is proposed to improve the incorporation of known information, which helps to improve the imputability of missing data in the model under different missing rates. Two real industrial processes are used to evaluate the effectiveness of the proposed method. The main contributions of this article are summarized as follows.

- 1) Position encoding is incorporated into DAE to explicitly account for the locations of the missing data. This enhancement allows the model to better understand and focus on the imputation of missing values.
- 2) The input variables are dynamically expanded to consider the autocorrelation of the missing variables. This approach aids in capturing the underlying patterns and relationship within the data.
- 3) An SPL framework is introduced to improve the incorporation of known information during model training. This framework gradually introduces more complex samples into the training process, enabling the model to learn in a more structured manner.
- 4) The PE-DAE+SPL method is designed effectively to handle varying rates of missing data. Unlike traditional models, which are often trained with a fixed missing rate, this approach adjusts to different missing-data scenarios, making it more robust and applicable to real-world industrial processes.

II. BACKGROUND

A. Denoising Autoencoder

The DAE is an unsupervised neural network, which seeks to reconstruct the clean input data from its randomly corrupted version. Fig. 1 shows the structure of a DAE, where the initial input $\mathbf{x}$ is the first corrupted by some stochastic mapping, such as additive Gaussian noise, masking noise (part of $\mathbf{x}$ is set to 0), or salt-and-pepper noise (part of $\mathbf{x}$ is set to 0 or 1). The corrupted $\tilde{\mathbf{x}}$ is then used as the input to the encoder, which transfers the corrupted $\tilde{\mathbf{x}}$ to the hidden representation $\mathbf{h}$ via the mapping function f , that is,

$$\mathbf{h} = f(\tilde{\mathbf{x}}) = s_f(\mathbf{W}\tilde{\mathbf{x}} + \mathbf{b}) \quad (1)$$

where $\mathbf{W} \in R^{d_h \times d_x}$ and $\mathbf{b} \in R^{d_h}$ are, respectively, the weight matrix and bias of the encoder and s_f is the activation function. $\mathbf{h}$ is then mapped back to the reconstructed input $\hat{\mathbf{x}}$ at the output layer using the mapping function g given as

$$\hat{\mathbf{x}} = g(\mathbf{h}) = s_g(\mathbf{W}\mathbf{h} + \hat{\mathbf{b}}) \quad (2)$$

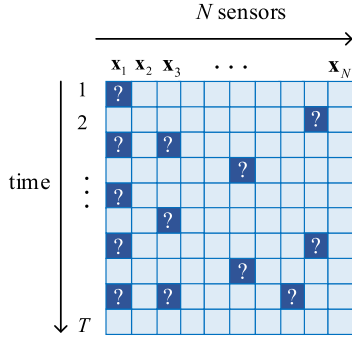


Fig. 2. Example of industrial process dataset with missing values.

where $\hat{\mathbf{W}} \in R^{d_x \times d_h}$ and $\hat{\mathbf{b}} \in R^{d_x}$ are, respectively, the corresponding weight matrix and bias vector and s_g is the activation function of the decoder.

The activation functions s_f and s_g are generally nonlinear activation functions. The parameter set for an autoencoder is given as $\theta = \{\mathbf{W}, \hat{\mathbf{W}}, \mathbf{b}, \hat{\mathbf{b}}\}$. Let the observed raw input dataset be $\mathbf{x}_{(t)} \in \{\mathbf{x}_{(1)}, \mathbf{x}_{(2)}, \dots, \mathbf{x}_{(T)}\}$, where T is the number of samples. The parameter set θ can be obtained by minimizing the following reconstruction error:

$$\begin{aligned} J(\theta) &= \frac{1}{2T} \sum_{t=1}^T L(\mathbf{x}_{(t)}, \hat{\mathbf{x}}_{(t)}) \\ &= \frac{1}{2T} \sum_{t=1}^T \|\mathbf{x}_{(t)} - \hat{\mathbf{x}}_{(t)}\|^2. \end{aligned} \quad (3)$$

B. DAE for Imputation

Missing values frequently arise from challenging working conditions or uncontrollable (intentional or unintentional) factors, such as sensor and communication failures or deliberate malicious cyberattacks. What makes matters worse is that prompt restoration is often impractical or entails excessive costs. This article focuses on predicting and imputing the missing data within datasets collected from industrial processes. Fig. 2 shows an example of an industrial dataset with missing values, where the color of the missing position is marked in a darker color. Assume that the process dataset is $\mathbf{X} \in R^{T \times N}$, where N represents the number of different sensors. It is obvious that the missing values in sensors $\mathbf{x}_1$ and $\mathbf{x}_3$ are caused by their lower sampling rate than other sensors, while the missing values in other sensors are missing at random.

Since the DAE cannot be trained on a dataset with the missing values, before using the DAE as a tool for data imputation, some additional steps are required. In general, the process of DAE-based imputation is given as follows.

- 1) Remove some values from the complete data $\mathbf{X}$ to form an incomplete training dataset and then divide dataset $\mathbf{X}$ into a training set $\mathbf{X}'$ and testing set $\mathbf{X}^n$.
- 2) Replace the missing values in the incomplete training dataset with the mean values of each sensor to form the corrupted dataset $\mathbf{X}'$.
- 3) Construct and learn the imputation model based on the corrupted dataset $\mathbf{X}'$ and learn the useful feature representations.

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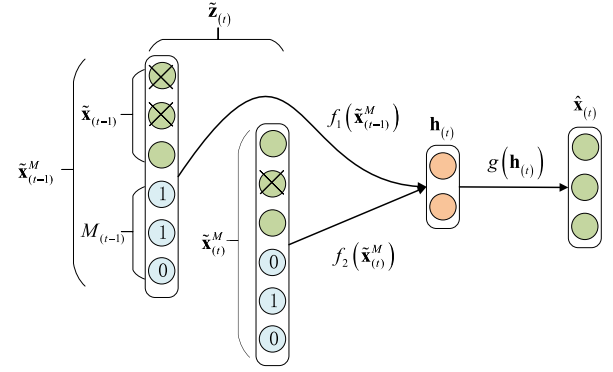


Fig. 3. Structure of PE-DAE (X-marked neurons represent missing data).

- 4) Similar to the training step, all the missing values in the testing set $\mathbf{X}^n$ are initialized to the mean of the nonmissing values for each variable. Then, it is input into the trained DAE model, which can directly estimate the missing values of the input vectors.

III. METHOD

A DAE is usually used to obtain complex and abstract feature representations that can improve the performance of the missing-data imputation [35]. However, the industrial application of the original DAE-based data imputation methods has some problems. First, the data in the industrial process are usually a time series, that is, the previous values of the process will affect future values of the process. However, the original DAE only uses the samples at the current moment for the data imputation. Second, the pre-imputation of missing data leads to the loss of missing location information, so it is difficult for DAE to determine which locations are missing data. Moreover, the definition of the loss function forces the DAE to focus on minimizing the overall reconstruction errors rather than the precise reconstruction of missing data. Finally, when training a DAE-based data imputation model, the training samples only randomly delete some data with a certain missing rate. Therefore, the trained model makes it difficult to generalize the different missing rates. Therefore, PE-DAE is proposed in this article, which changes the network structure, loss function, and training method of a DAE. These changes help the network to find the internal structure in the missing data and reduce imputation errors.

In this article, the missing values are represented as “NaN,” and a matrix $M \in R^{T \times N}$ is used to represent the missing position information. The elements in M are defined as

$$M_{i(t)} = \begin{cases} 1, & \mathbf{x}_{i(t)} = \text{NaN} \\ 0, & \text{otherwise} \end{cases} \quad (4)$$

where $\mathbf{x}_{i(t)}$ is the t th data sample from the i th sensor, $i = 1, 2, \dots, N$, and $t = 1, 2, \dots, T$.

A. Position-Encoding DAE

Fig. 3 shows a schematic of PE-DAE, which is a modification of the original DAE. PE-DAE learns the latent variable $\mathbf{h}_{(t)}$ from the missing data and then reconstructs the estimate $\hat{\mathbf{x}}_{(t)}$ of the missing data from $\mathbf{h}_{(t)}$. The original input with

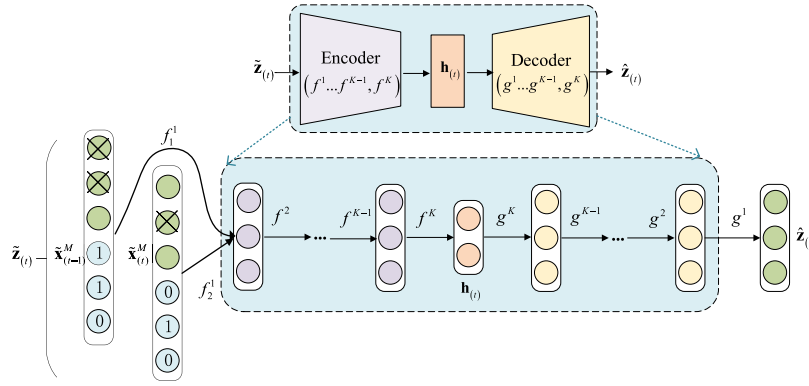


Fig. 4. Structure of deep PE-DAE.

missing data at the time $t - 1$ is $\tilde{\mathbf{x}}_{(t-1)}$, while $M_{(t-1)}$ is the positional encoding of the same dimension as $\tilde{\mathbf{x}}_{(t-1)}$. The input vector that concatenates $\tilde{\mathbf{x}}_{(t-1)}$ and $M_{(t-1)}$ together is given as $\tilde{\mathbf{x}}_{(t-1)}^M$ so that $\tilde{\mathbf{x}}_{(t-1)}^M$ contains the information of the missing position. Similarly, the input vector at the current time t can be expressed as $\tilde{\mathbf{x}}_{(t)}^M$. After obtaining $\tilde{\mathbf{x}}_{(t-1)}^M$ and $\tilde{\mathbf{x}}_{(t)}^M$, they simultaneously serve as the final input $\tilde{\mathbf{z}}_{(t)}$ to PE-DAE. The functions f and g are still used to represent the mapping function of, respectively, the encoder and the decoder. Since the encoder is divided into two parts, f_1 and f_2 , are used to represent the mapping function at the previous time and the current time, respectively. This can be formulated as

$$\begin{aligned} \tilde{\mathbf{z}}_{(t)} &= \begin{bmatrix} \tilde{\mathbf{x}}_{(t-1)}^M \\ \tilde{\mathbf{x}}_{(t)}^M \end{bmatrix}, \quad \tilde{\mathbf{x}}_{(t-1)}^M = \begin{bmatrix} \tilde{\mathbf{x}}_{(t-1)} \\ M_{(t-1)} \end{bmatrix}, \quad \tilde{\mathbf{x}}_{(t)}^M \\ &= \begin{bmatrix} \tilde{\mathbf{x}}_{(t)} \\ M_{(t)} \end{bmatrix}, \quad f = \begin{bmatrix} f_1 \\ f_2 \end{bmatrix}. \end{aligned} \quad (5)$$

Forward propagation can be written as

$$\hat{\mathbf{x}}_{(t)} = g(\mathbf{h}_{(t)}) = g(f(\tilde{\mathbf{z}}_{(t)})) = g \left(\begin{bmatrix} f_1(\tilde{\mathbf{x}}_{(t-1)}^M) \\ f_2(\tilde{\mathbf{x}}_{(t)}^M) \end{bmatrix} \right) \quad (6)$$

where $\hat{\mathbf{x}}_{(t)}$ is an estimate of the samples with missing values and should be as close as possible to the label value $\mathbf{x}_{(t)}$.

The learning goal of the original DAE is to minimize the loss function, so as to achieve a small overall reconstruction error. However, the data imputation is more concerned with the reconstruction of missing values. A reasonable reconstruction should be more accurate for those missing values. Therefore, for the training of PE-DAE, an unequal loss function is designed, which focuses more on achieving accurate reconstruction for missing values. Consequently, PE-DAE can be trained to decrease the following unequal reconstruction errors:

$$J_{\text{PE-DAE}}(\theta) = \frac{1}{2T} \sum_{t=1}^T \sum_{i=1}^N m_i \|\mathbf{x}_{i(t)} - \hat{\mathbf{x}}_{i(t)}\|^2 \quad (7)$$

$$m_{i(t)} = (1 - M_{i(t)}) + \alpha M_{i(t)} \quad (8)$$

where $m_{i(t)}$ represents an attention coefficient. For nonmissing values, it takes a value of 1, whereas for the missing values, it exceeds 1; $\alpha \geq 1$ is determined through trial and error.

The enhanced performance of the missing-data imputation is facilitated by the acquisition of higher level representations

through deep neural networks [36]. Therefore, the capabilities of PE-DAE are extended by augmenting the number of hidden layers, which results in a deeper architectural configuration. The training procedure for the deep PE-DAE is shown in Fig. 4. In the deep PE-DAE, incomplete data $\tilde{\mathbf{z}}$ are sequentially input into the encoder part in a chronological order. Then, the encoder ($f^1, \dots, f^{K-1}, f^K$) encodes the incomplete data into a feature representation $\mathbf{h}$. It uses neurons in the latent space to represent incomplete data to alleviate the need for explicitly modeling essential relationships between variables. Therefore, deep architecture can learn the representations and complex relationships of different variables for missing-data imputations [37]. Finally, the feature representation $\mathbf{h}$ is decoded into complete data $\hat{\mathbf{z}}$ by the decoder ($g^1, \dots, g^{K-1}, g^K$). Unlike shallow PE-DAE, in this process, the encoder and decoder are replaced with multilayer neural networks.

B. Self-Paced Learning

Most deep-learning-based missing-data imputation methods use either complete data or data with fixed missing rates during the training phase to train the network. However, the actual missing rate in real-world data is unknown, which can result in inadequate generalizability of the model. Moreover, it will also lead to limited training samples available in the case of high missing rates. A reasonable training method should consider the rational use of incomplete samples rather than simply discarding them. Therefore, a new training method called SPL is proposed to solve these problems. SPL uses sample network training with different missing rates, which allows incomplete data with different missing rates to participate in the network training. The overall process, shown in Fig. 5, can be divided into the training and testing steps.

1) In the training step, the first step is to generate training datasets with different missing rates. In Fig. 5, missing values in the dataset have been marked in a darker color, with $\tilde{\mathbf{z}}_{(10\%)}$, $\tilde{\mathbf{z}}_{(20\%)}$, $\tilde{\mathbf{z}}_{(30\%)}$, and $\tilde{\mathbf{z}}_{(40\%)}$ representing the training datasets with missing rates of, respectively, 10%, 20%, 30%, and 40%. Next, the training datasets are arranged in ascending order based on the missing rates and sequentially used as input for the PE-DAE encoder. This means that the number of training samples for PE-DAE increases fourfold. Therefore, PE-DAE can learn more abundant information about the missing-value imputation. Furthermore, organizing the missing

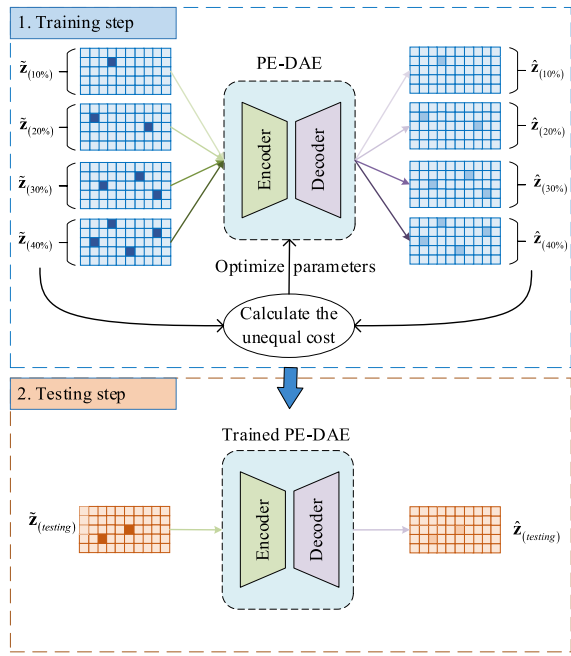


Fig. 5. SPL method.

rates from low to high allows the model to learn from simple to complex scenarios. This arrangement can significantly enhance both learning effectiveness and speed. The PE-DAE's encoder can handle the complex relationship between variables by mapping the incomplete data to the latent feature space. Next, the PE-DAE's decoder can reconstruct these latent variables back to the complete data to obtain the imputation result $\hat{\mathbf{z}}_{(10\%)}$, $\hat{\mathbf{z}}_{(20\%)}$, $\hat{\mathbf{z}}_{(30\%)}$, and $\hat{\mathbf{z}}_{(40\%)}$. Finally, the unequal cost function between $\tilde{\mathbf{z}}$ and $\hat{\mathbf{z}}$ is calculated using (7). All the network parameters are optimized using iterative updating. The trained PE-DAE is obtained after the termination condition is satisfied.

2) In the testing step, testing data with missing values is input to the trained PE-DAE. The output corresponding to the missing values is taken as the final estimation.

SPL maximizes the use of the information from the complete dataset. Moreover, considering networks trained by certain data with a fixed missing rate often leads to large generalization errors (such as the generalization errors caused by the difference in missing rates between the training set and the testing set). With the participation of the large sets of training data and the learning going further, PE-DAE+SPL will deeply learn the essential characteristics of the data, which is conducive to the missing-data imputation.

IV. CASE STUDIES

The missing-data completion performance of PE-DAE+SPL is evaluated by the debutanizer column and the hydrocracking process. The code of PE-DAE+SPL can be found at <https://github.com/xiaochenzi94/PE-DAE-SPL>.

A. Experimental Setup

1) *Algorithms Compared:* To evaluate the performance of the PE-DAE+SPL method, the five mainstream missing-data imputation methods were selected for comparison, namely,

improved k -nearest-neighbor (KNN)-based imputation [38], autoencoder (AE)-based imputation, DAE, generative-adversarial-network (GAN)-based imputation [39], and bidirectional recurrent imputation for time series (BRITS) [34]. The optimal hyperparameters for each method are selected from set ranges. The nearest neighbors of KNN are selected from 1 to 5. For AE, DAE, and PE-DAE, the number of hidden layers is selected from 1 to 5. The number of neurons for each hidden layer is chosen in the range [40, 100]. As for GAN, the network structures of the generator and discriminator are the same, for all three layers. The activation function is sigmoid. For the minibatch gradient descent, the batch size is chosen from the range [32, 64], the epoch is chosen from the range [800, 1200], and the learning rate is chosen from the range [0.01, 0.1]. For BRITS, the dimension of the hidden state is chosen from the range [32, 64, 128]. For the proposed algorithm, the ratio α is set to 1.5.

2) *Evaluation Criteria:* Root-mean-squared error (RMSE) and mean absolute percentage error (MAPE) are chosen as measures of imputation performance in this study, which are calculated using

$$\text{RMSE} = \sqrt{\frac{\sum_{n=1}^N (\mathbf{y}_n - \hat{\mathbf{y}}_n)^2}{N-1}} \quad (9)$$

$$\text{MAPE} = \frac{1}{N} \sum_{n=1}^N \left| \frac{\mathbf{y}_n - \hat{\mathbf{y}}_n}{\mathbf{y}_n} \right| \times 100\% \quad (10)$$

where $\hat{\mathbf{y}}_n$ is the estimated missing value, $\mathbf{y}_n$ is the actual value, and N is the number of missing values. To avoid the problem that the denominator of the MAPE is zero or close to zero, all the data are normalized to lie between 0.1 and 0.9.

B. Debutanizer Column Process

The debutanizer column plays a crucial role in the refining process, particularly in the desulfurization and naphtha-separation unit, as shown in Fig. 6. One of its main tasks is to reduce the butane (C_4) content at the bottom of the debutanizer [40]. Seven sensors were used to monitor signals such as temperature, flow, and pressure. The debutanizer column dataset is a widely used benchmark in data-driven modeling [41]. In this study, 2394 time series samples were used to evaluate the performance of the proposed algorithm, of which the first 1500 samples were used as the training set and validation set, and the remaining 894 samples as the testing set. The missing values were set to NaN. The missing rates are set as 10%, 20%, 30%, 40%, and 50%.

First, the impact of parameter selection for the various methods for the debutanizer column process dataset was analyzed based on the RMSE values. Fig. 7 shows the performance of some of the model parameters on the debutanizer process dataset. It can be seen that the number of nearest neighbors has a slight impact on the RMSE values of KNN. For GAN, experimental results show that the performance of GAN does not get better as the number of hidden-layer neurons increases, which may be caused by overfitting problems. For DAE, the number of hidden layers significantly influences the experimental results. Excessive hidden layers, on the other hand,

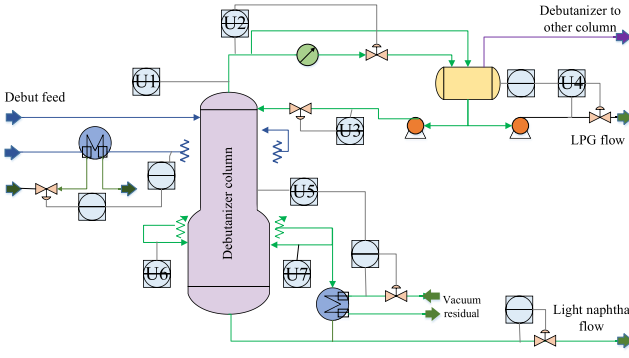


Fig. 6. Flowchart of the debutanizer column.

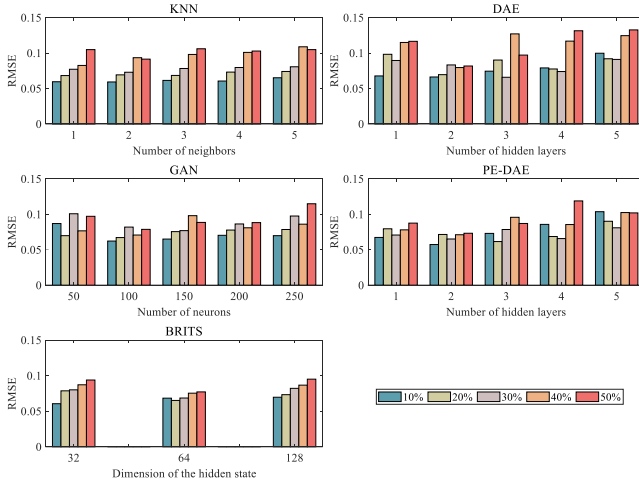


Fig. 7. Impact of the hyperparameters for the debutanizer column dataset.

tend to result in a decline in performance. For BRITS, medium dimensions (e.g., 64) provide enough capacity to model moderately complex patterns. In summary, the parameters in these algorithms are optimized and selected based on the RMSE of the debutanizer column dataset, as shown in Table I.

The experimental results for the seven models are summarized in Table II, in which the RMSE and MAPE values are the mean values based on the five replicate experiments. The best result is bolded and underlined, while the second-best result is simply bolded. It can be observed that the imputation performance of PE-DAE+SPL outperforms the other methods for different missing rates. The second-best results are all from the PE-DAE and DAE+SPL. Of the five methods KNN, DAE, GAN, BRITS, and PE-DAE without the SPL training strategy, PE-DAE achieved the lowest RMSE and MAPE at different missing rates, which implies that the implemented position encoding and dynamic expansion techniques improve the imputation performance of DAE. Furthermore, based on the results of DAE and DAE+SPL, the RMSE of DAE+SPL is lower than DAE by 14.20%, 8.35%, 7.65%, 12.70%, and 10.65%. Similar observations can be found in PE-DAE and PE-DAE+SPL, the RMSE of PE-DAE+SPL is lower than PE-DAE by 17.98%, 17.56%, 16.28%, 12.80%, and 11.34%. The above results show that when SPL learns the network by gradually moving from easy to complex missing samples in training, the accuracy of imputation is significantly improved.

TABLE I
OPTIMAL PARAMETERS SELECTED FOR DIFFERENT METHODS

Dataset	Missing rates	KNN	DAE	GAN	BRITS	PE-DAE
Debutanizer column	10%	2	2	100	32	2
	20%	1	2	100	64	3
	30%	2	3	200	64	2
	40%	1	2	100	64	2
	50%	2	2	100	64	2

Of note, the estimation curve of PE-DAE+SPL at 20% missing rate is shown in Fig. 8, which gives the estimated results of seven variables for the debutanizer column. In the subview of each variable, the real and estimated values are indicated in, respectively, red and blue. It can be seen that the trend of the estimated values remains consistent with the real values in most cases. The results around the 100th sample for variable 2 are relatively poor because of the large fluctuations in the true value. This can be due to equipment failure or inaccurate measurements.

Fig. 9. shows a comparative analysis of the correlation between estimated values and actual values for KNN, GAN, DAE, BRITS, and PE-DAE+SPL at a missing rate of 20%. If the model accurately estimates a missing value, the corresponding sample point lies precisely on the diagonal. Therefore, the closer the sample points are to the diagonal, the more accurate the imputation results. It can be seen that the PE-DAE+SPL estimation results are more concentrated near the diagonal, which indicates that PE-DAE+SPL exhibits better interpolation efficacy compared with the other methods.

The test results of different SPL strategies during training on the debutanizer are summarized in Table III. In this context, A1 denotes a training set comprising only one missing rate, $\tilde{\mathbf{z}}_{(10\%)}$; A2 denotes a training set containing two missing rates, $[\tilde{\mathbf{z}}_{(10\%)}, \tilde{\mathbf{z}}_{(20\%)}]$; A3 denotes three missing rates, $[\tilde{\mathbf{z}}_{(10\%)}, \tilde{\mathbf{z}}_{(20\%)}, \tilde{\mathbf{z}}_{(30\%)}]$; A4 denotes four missing rates, $[\tilde{\mathbf{z}}_{(10\%)}, \tilde{\mathbf{z}}_{(20\%)}, \tilde{\mathbf{z}}_{(30\%)}, \tilde{\mathbf{z}}_{(40\%)}]$; and A5 denotes five missing rates, $[\tilde{\mathbf{z}}_{(10\%)}, \tilde{\mathbf{z}}_{(20\%)}, \tilde{\mathbf{z}}_{(30\%)}, \tilde{\mathbf{z}}_{(40\%)}, \tilde{\mathbf{z}}_{(50\%)}]$. The phrase “Change (%)” means the change relative to A1 in percent. It can be seen that the RMSE keeps decreasing as the SPL training progresses but does not decrease the RMSE significantly after A3. Therefore, the SPL extends the training set with various missing rates, facilitating the model’s learning from simple to complex, thereby effectively reducing generalization errors. It is worth noting that A5 shows only a slight improvement compared to A4, indicating that as the variety of missing-data types increases, the magnitude of performance improvement gradually diminishes.

C. Hydrocracking Process

Hydrocracking process is an essential part of the petrochemical industry. In this process, crude oil is cracked and converted into light molecular oils such as gasoline, kerosene, and diesel. The flowchart of a hydrocracking process is shown in Fig. 10. It consists of four parts: feeding, reaction, high- and low-pressure separation, and fractionation [42]. In this process, 34 process variables such as temperature, pressure, and flow

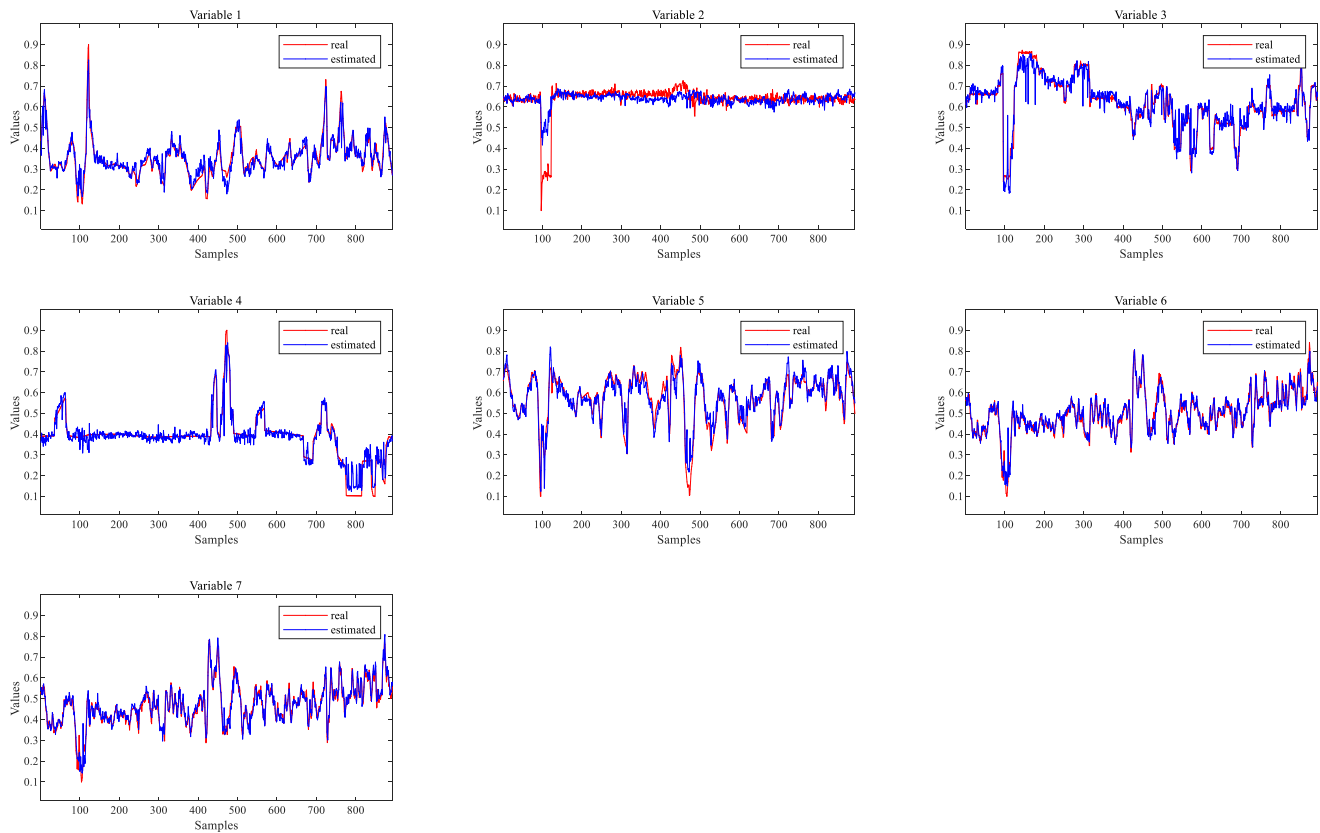


Fig. 8. Estimation curve of PE-DAE+SPL at 20% missing rate of the debutanizer column.

TABLE II
EXPERIMENTAL RESULTS OF DEBUTANIZER COLUMN

Missing rates	RMSE							MAPE						
	KNN	DAE	DAE+SPL	GAN	BRITS	PE-DAE	PE-DAE + SPL	KNN	DAE	DAE+SPL	GAN	BRITS	PE-DAE	PE-DAE + SPL
10%	0.0592	0.0662	0.0568	0.0622	0.0607	0.0573	<u>0.0470</u>	10.04	11.60	9.54	10.75	10.01	9.56	<u>7.67</u>
20%	0.0682	0.0695	0.0637	0.0671	0.0651	0.0615	<u>0.0507</u>	11.90	11.98	10.68	11.53	10.77	10.36	<u>8.32</u>
30%	0.0729	0.0732	0.0676	0.0769	0.0685	0.0651	<u>0.0545</u>	12.31	12.36	11.00	13.11	10.96	10.71	<u>8.41</u>
40%	0.0825	0.0795	0.0694	0.0707	0.0753	0.0711	<u>0.0620</u>	14.58	13.69	10.96	11.82	12.34	11.92	<u>9.48</u>
50%	0.0914	0.0817	0.0730	0.0787	0.0771	0.0732	<u>0.0649</u>	17.04	14.33	12.05	13.62	13.15	12.46	<u>10.46</u>

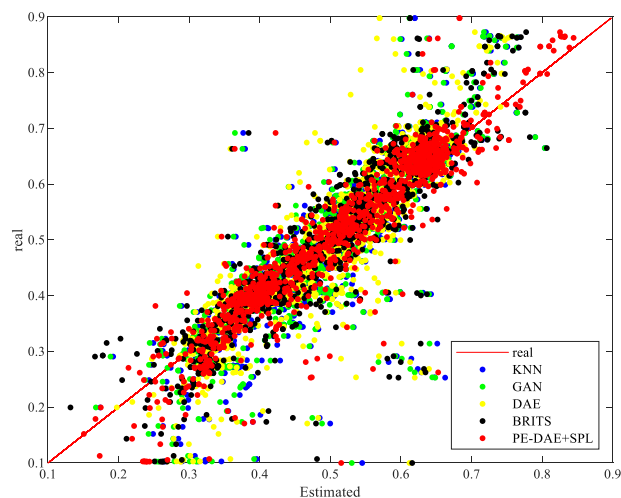


Fig. 9. Correlation between the estimated values and real values for a missing rate of 20% in the debutanizer column.

rate that are assumed to be important for product-quality assessment were selected for model training and testing [43].

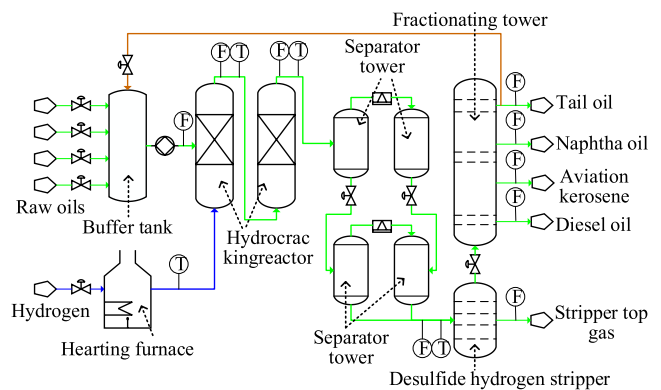


Fig. 10. Flowchart of the hydrocracking process.

There are 1921 samples collected from the hydrocracking process, of which the first 1500 samples were used as the training and validation sets, while the remaining 421 samples were used as the testing set. The missing rates are set as 10%, 20%, 30%, 40%, and 50%.

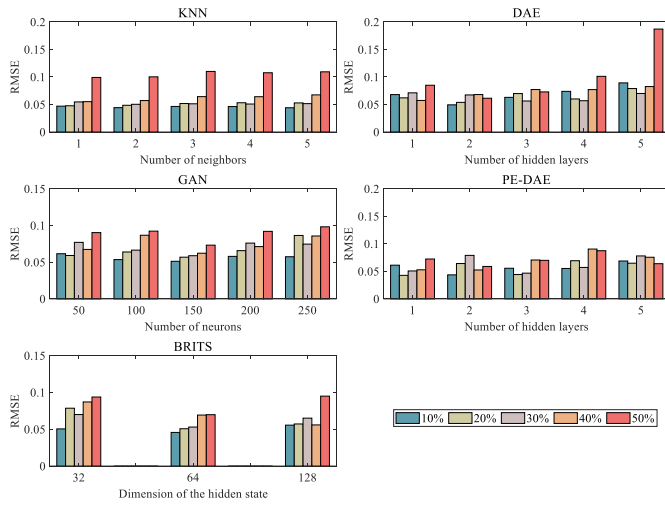


Fig. 11. Impact of the parameters for the hydrocracking process dataset.

Fig. 11 shows the impact of different parameters on the hydrocracking dataset. Overall, the influence of these parameters on the results is similar to that observed in the debutanizer column process dataset. The parameters for the algorithms are optimized and selected based on the RMSE of the hydrocracking process dataset, as shown in Table IV.

Table V shows the RMSE for the testing set using different SPL strategies during training. The phrase “Change (%)” means the change relative to A1 in percent. It can be observed that, with the progression of SPL training, the overall trend of RMSE is decreasing. However, it is worth noting that having more types of missing data does not necessarily lead to better results. The RMSE for A5 is unexpectedly higher than that of A4 at 30% and 50% missing rates. This may be attributed to potential overfitting caused by an excessive number of training set samples. Therefore, it is crucial not to excessively increase the number of training samples in SPL.

The experimental results for the seven models are summarized in Table VI, in which the RMSE and MAPE values are the mean values based on five replicate experiments. From the results, it can be observed that the RMSE and MAPE of PE-DAE are smaller than those of KNN, DAE, GAN, and BRITS. This is attributed to the incorporation of missing position information and the autocorrelation information of missing variables in PE-DAE. In addition, it can be seen that for different missing rates, the RMSE and MAPE of PE-DAE+SPL are the smallest, suggesting its better performance in handling missing data. Of these, the RMSE and MAPE of PE-DAE+SPL are lower than the PE-DAE. This suggests that the PE-DAE model, enhanced with SPL, performs well in accurately imputing missing values across a range of missing rates.

In Fig. 12, the estimation curve for the first nine variables in the hydrocracking with a missing rate of 20% for PE-DAE+SPL is shown. It can be seen that the estimates provided by PE-DAE+SPL closely approximate the real values, with only a tiny percentage of the estimated values deviating.

TABLE III
TEST RESULTS OF DIFFERENT SPL STRATEGIES ON
THE DEBUTANIZER COLUMN

Missing rates	Metric	A1	A2	A3	A4	A5
10%	RMSE	0.0573	0.0516	0.0495	0.0478	0.047
	Change (%)	0	-9.95	-13.61	-16.58	-17.98
20%	RMSE	0.0615	0.0561	0.0534	0.0519	0.0507
	Change (%)	0	-8.78	-13.17	-15.61	-17.56
30%	RMSE	0.0651	0.0601	0.0577	0.0558	0.0545
	Change (%)	0	-7.68	-11.37	-14.29	-16.28
40%	RMSE	0.0711	0.0665	0.0651	0.0627	0.062
	Change (%)	0	-6.47	-8.44	-11.81	-12.80
50%	RMSE	0.0732	0.0688	0.068	0.0655	0.0649
	Change (%)	0	-6.01	-7.10	-10.52	-11.34

TABLE IV
OPTIMAL PARAMETER VALUES FOR THE HYDROCRACKING DATASET

Dataset	Missing rates	KNN	DAE	GAN	BRITS	PE-DAE
Hydrocracking process	10%	2	2	100	64	2
	20%	1	2	200	64	3
	30%	2	3	200	64	3
	40%	1	1	200	128	2
	50%	1	2	200	64	2

TABLE V
TEST RESULTS OF DIFFERENT SPL STRATEGIES ON
THE HYDROCRACKING PROCESS

Missing rates	Metric	A1	A2	A3	A4	A5
10%	RMSE	0.0436	0.0385	0.0385	0.0384	0.0368
	Change (%)	0	-11.70	-11.70	-11.93	-15.60
20%	RMSE	0.0442	0.0424	0.0415	0.0396	0.039
	Change (%)	0	-4.07	-6.11	-10.41	-11.76
30%	RMSE	0.0467	0.0435	0.0426	0.0397	0.0411
	Change (%)	0	-6.85	-8.78	-14.99	-11.99
40%	RMSE	0.0525	0.048	0.0457	0.0431	0.0425
	Change (%)	0	-8.57	-12.95	-17.90	-19.05
50%	RMSE	0.0589	0.0531	0.0518	0.0451	0.0457
	Change (%)	0	-9.85	-12.05	-23.43	-22.41

Fig. 13 shows a scatter plot of the imputed output and the real values for the selected methods using the hydrocracking dataset with a missing rate of 20%. It can be seen that PE-DAE+SPL has a greater number of data sample points that are closer to the diagonal, indicating the superiority of PE-DAE+SPL compared to the other four methods.

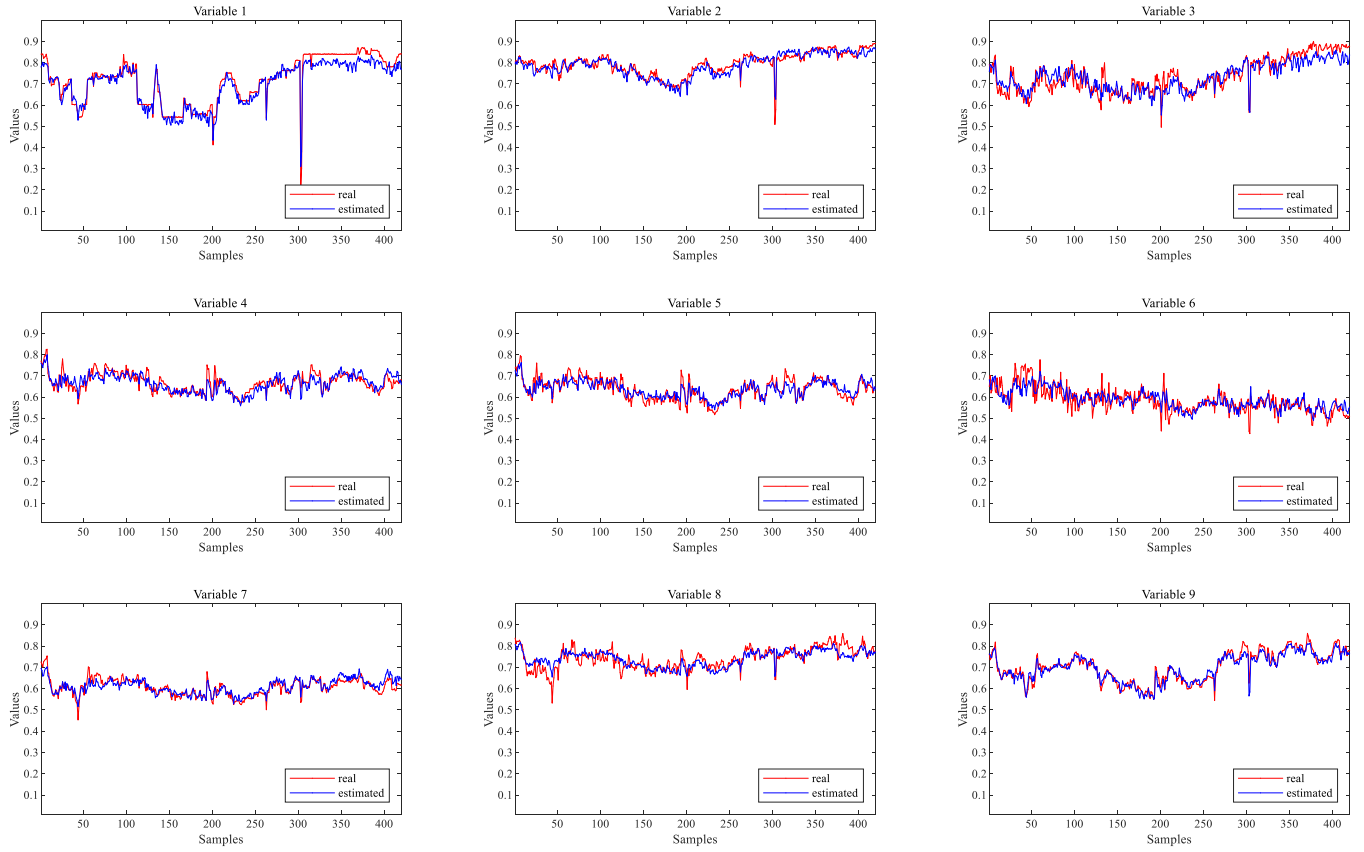


Fig. 12. Estimation curve of PE-DAE+SPL at 20% missing rate of the hydrocracking process.

TABLE VI
EXPERIMENTAL RESULTS FOR THE HYDROCRACKING DATASET

Missing rates	RMSE							MAPE						
	KNN	DAE	DAE+SPL	GAN	BRITS	PE-DAE	PE-DAE + SPL	KNN	DAE	DAE+SPL	GAN	BRITS	PE-DAE	PE-DAE + SPL
10%	0.0443	0.0494	0.0448	0.0512	0.0457	0.0436	<u>0.0368</u>	5.62	6.47	5.78	6.76	5.98	5.57	<u>4.71</u>
20%	0.0477	0.0539	0.0454	0.0568	0.0507	0.0442	<u>0.0390</u>	6.18	7.04	5.88	7.40	6.65	5.66	<u>4.90</u>
30%	0.0504	0.0564	0.0459	0.0588	0.0530	0.0467	<u>0.0411</u>	6.55	7.36	6.01	7.76	7.10	6.11	<u>5.12</u>
40%	0.0552	0.0574	0.0467	0.0622	0.0559	0.0525	<u>0.0425</u>	7.23	7.34	6.17	8.26	7.28	6.88	<u>5.67</u>
50%	0.0992	0.0614	0.0496	0.0732	0.0698	0.0589	<u>0.0451</u>	13.01	8.43	6.51	9.73	8.97	8.08	<u>5.85</u>

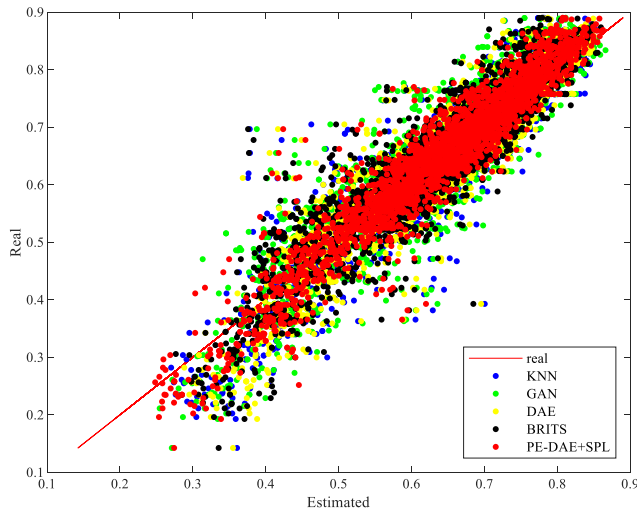


Fig. 13. Correlation between the estimated values and real values at a 20% missing rate in the hydrocracking process.

V. CONCLUSION

In this article, a PE-DAE is proposed for missing value imputation. Building upon the traditional DAE, the input structure of PE-DAE is redesigned to incorporate information about the positions of missing values while also accounting for the autocorrelation of the missing variables. In addition, SPL is proposed, which trains the network on data with varying missing rates sequentially from the simplest case with low missing rates to the most challenging with high missing rates. The imputation performance of PE-DAE+SPL is validated using examples from two chemical processes, the debutanizer column and the hydrocracking process. The results show that PE-DAE outperforms KNN, DAE, GAN, and BRITS in imputation. Furthermore, the results show that networks trained with the proposed SPL have higher imputation accuracy compared to those trained with the traditional training methods. PE-DAE+SPL has a lower RMSE and fewer MAPE. In most cases, the estimates provided by

PE-DAE+SPL can capture the trends of the true value curves. Despite the promising results, the study's limitations include the difficulty of selecting optimal parameters for SPL. Future research should focus on developing automated and efficient methods for SPL parameter tuning.

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